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Semester Project 3 - 2024

[Provide Group Members contributions in %, only if not equal]

[Short history of the **Device** in your name, max. ½ page]

Abstract

[The **Abstract** of a technical report is a concise summary that provides an overview of the entire report. It typically includes the following key elements:

1. **Purpose of the Project**
 - State the problem the project addresses.
 - Mention the motivation or context for undertaking the work.
2. **Objectives**
 - Briefly describe the main goals of the project.
3. **Methods or Approach**
 - Highlight the methodology or techniques used to solve the problem.
 - Mention any tools, technologies, or frameworks employed.
4. **Key Results or Findings**
 - Summarize the most important outcomes or achievements of the project.
 - Mention whether the goals were met and what was accomplished.
5. **Conclusion**
 - Provide a high-level reflection on the impact or significance of the work.
 - Optionally, hint at future possibilities or improvements.

The Abstract should be:

- **Brief:** Typically, 150–250 words.
- **Self-contained:** Readers should understand the project without reading the rest of the report.
- **Informative:** Focus on what was done, how, and the results.
- **Non-technical:** Avoid excessive jargon or technical details; it should be understandable by a broader audience.

It should be written last, after completing the report, to accurately reflect its content.]

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1. Introduction (Oskar)

[Provides an overview of the project, including the context, objectives, scope, and importance. This includes the desk use case as well the 3rd semester structure. It explains the problem being addressed and sets the stage for the subsequent chapters.]

1.1. Motivation

[Lorem ipsum]

1.2. Objective

[Lorem ipsum]

1.3. Delimitation

[After introducing the project and its objectives, include a paragraph explaining:

- What the project will not address (e.g., specific features, areas, or use cases that are out of scope)
- Constraints or limitations (e.g., time, budget, technology, or resources)]

2. Methodology (Oskar)

- Scrum
- Gantt chart
-

[Describes the approach and techniques employed to address the problem. This could include research methods, tools, technologies, and frameworks used during the project and the 3rd semester courses.

You build the project based on the experience and knowledge accumulated during the previous semesters—don't forget that.]



Figure 1 An example figure

2.1. [Sub-section]

[Lorem ipsum]

2.1.1. [Sub-sub-section]

[Lorem ipsum]

3. Problem Analysis (Oskar)

[A detailed breakdown of the problem being tackled. This section analyses the key challenges, potential bottlenecks, and constraints. It may also include a review of related work or literature.]

3.1. [Sub-section]

[Lorem ipsum]

3.1.1. [Sub-sub-section]

[Lorem ipsum]

4. Requirements (Mathias)

[Lists the functional and non-functional requirements for the project, clearly stating what the system, application, or solution is expected to achieve.]

Table 1 An example table

		X
	X	
X		

4.1. [Sub-section]

[Lorem ipsum]

4.1.1. [Sub-sub-section]

[Lorem ipsum]

5. Design (Oskar)

[Outlines the architecture, system design, and key components of the project. This chapter may include diagrams like UML models, flowcharts, or block diagrams to visually represent the design.]

5.1. [Sub-section]

[Lorem ipsum]

5.1.1. [Sub-sub-section]

[Lorem ipsum]

6. Implementation (Mathias)

[Documents the process of developing the solution, including key algorithms, code snippets, libraries, and tools used. Focus is on the technical realization of the design.]

6.1. [Sub-section]

[Lorem ipsum]

6.1.1. [Sub-sub-section]

[Lorem ipsum]

7. Validation (Mathias)

[Details the testing and validation processes to ensure the system meets the specified requirements. Includes test cases, results, and analysis of the performance or functionality of the solution.]

7.1. [Sub-section]

[Lorem ipsum]

7.1.1. [Sub-sub-section]

[Lorem ipsum]

8. Conclusion (Mathias)

[Summarizes the outcomes of the project, lessons learned, and areas for future improvement or further study. It may also reflect on how well the objectives were achieved.]

8.1. Summary

[Lorem ipsum]

8.2. Future Work

[Lorem ipsum]

References

There are no sources in the current document.

[This chapter includes a list of all sources cited throughout the report. In technical writing, adhering to a standardized referencing style ensures consistency, accuracy, and professionalism.]

What to Include:

- Research papers, journal articles, or books referenced during the project
- Online resources such as technical blogs, official documentation, or standards
- Any tools, libraries, or datasets that are cited

The **IEEE** style is commonly used in technical and engineering fields. It is a **numeric citation style** where references are numbered in the order they are cited in the text.

Key Features of IEEE Style:

1. In-Text Citations:
 - Use numbers in square brackets to cite sources, e.g., "As described in [1]..."
 - The numbers correspond to the order of citation, not alphabetical order
2. Reference List Format:
 - The references are listed at the end of the document, numbered in the same order as they appear in the text.
 - Each reference follows a specific format depending on the type of source (book, article, website, etc.).
3. Common Formats in IEEE Style:
 - Book: [1] Author(s), Title, Edition (if applicable). City: Publisher, Year.
 - Example: [1] A. B. Smith and C. D. Jones, Introduction to Embedded Systems, 3rd ed. New York: Springer, 2018.
 - Journal Article: [2] Author(s), "Title of article," Journal Name, vol. X, no. Y, pp. ZZZ-ZZZ, Month Year.
 - Example: [2] J. K. Doe, "An efficient algorithm for data compression," IEEE Transactions on Computers, vol. 67, no. 3, pp. 456-468, Mar. 2022.
 - Conference Paper: [3] Author(s), "Title of paper," in Proceedings of the Conference Name, Location, Year, pp. ZZZ-ZZZ.
 - Example: [3] K. L. Brown, "Innovations in autonomous systems," in Proceedings of the 15th International Conference on Robotics, Tokyo, Japan, 2021, pp. 34-45.
 - Website: [4] Author(s) (if available), "Title of web page," Website Name, Date of publication. [Online]. Available: URL
 - Example: [4] T. Nguyen, "Understanding neural networks," AI Research Blog, Jan. 2023. [Online]. Available: <https://www.aiblog.com/neural-networks>

Microsoft **Word** has built-in tools for managing references and citations, and it supports several citation styles, including IEEE.]

Appendix A

[The Appendix sections in the report is used to include supplementary information that is too detailed or extensive to be included in the main chapters but is still important for understanding the project.

Common appendices:

- Raw data or detailed calculations
- Code snippets or algorithms
- Detailed figures, graphs, or tables
- Technical specifications
- Test cases and results
- Meeting minutes or project logs
- User manuals or guides
- Legal documentation, agreements]

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